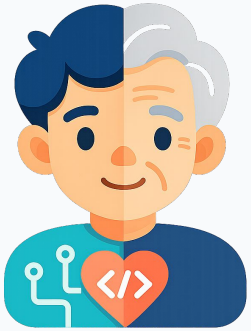
Turning Code into Compassion

A Visualize Aging Tool for Awareness and Reflection (AVATAR)



STUDENT RESEARCHERS

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SUPPORTED BY

NCSA SPIN • Fiddler Innovation Undergraduate Fellowship

(dis)ability DESIGN STUDIO

ILLINOIS
NCSA | National Center for Supercomputing Applications

PROBLEM

Young adults treat their future selves like strangers.

So today’s habits feel disconnected from long-term health.

OPPORTUNITY

Seeing a digitally aged version of yourself narrows that gap.

- More empathy
- More long-term planning

Owen et al., 2019; Dos Santos et al., 2025

INTRODUCTION

AVATAR

AVATAR shows young adults an image of their older selves, and explains why they age that way.

- Face aging tool
- Personalized visualization
- Future self simulation



KEY INSIGHT

Aging goes beyond the skin

Every layer of the face changes, not just the surface.

It varies with gender, ethnicity, environment, and genetics.

Skin	Wrinkles form and elasticity drops
Fat	Volume thins and shifts downward
Muscle	Tone and position change
Bone	Structure slowly reshapes

Hussein et al., 2025

Swift et al.

DIFFERENCE

Existing tools vs. AVATAR

	EXISTING TOOLS	AVATAR
Scope	One risk factor, such as smoking or sun	Genetics, environment, and multiple lifestyle habits together
Representation	Generic, exaggerated filters that fail on diverse skin tones	Identity-preserving AI grounded in biomedical data
Explanation	“Black Box” with no reasons given	Explains which features changed, which habits contributed, and why

HOW WE BUILT AVATAR

Summer

1 Research
Reviewed 60 sources on facial aging and existing simulators

2 Define
Chose what to model: age, demographics, smoking, UV, alcohol, stress, and diet

3 Generate
Generate a future self image with Diffusion-Based Editing

Fall

1 Research
Expand the scope of aging feature data and building a knowledge graph

2 IRB
Submitted IRB so we can test AVATAR with real users

3 Redesign
Rebuilt the aging pipeline with local mesh warping and AI-assisted rendering

Spring

1 Pilot Study
Run a pilot with 100 UIUC students and gather feedback

2 Transparency
Explain each aging features on the software: what changed, which habits caused it, and why

3 Improvements
Improve image accuracy with study feedback, new research, and better AI models

SUMMER SHOWCASE

STEM Career Exploration and Symposium





July 30, 2026

HOW AVATAR HAS EVOLVED

LAST LIGHTNING TALK

One aged image



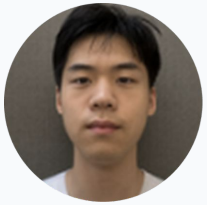
Input



Aged Results

END OF SUMMER

Aging that responds to habits



Input, age 20



Baseline, age 65



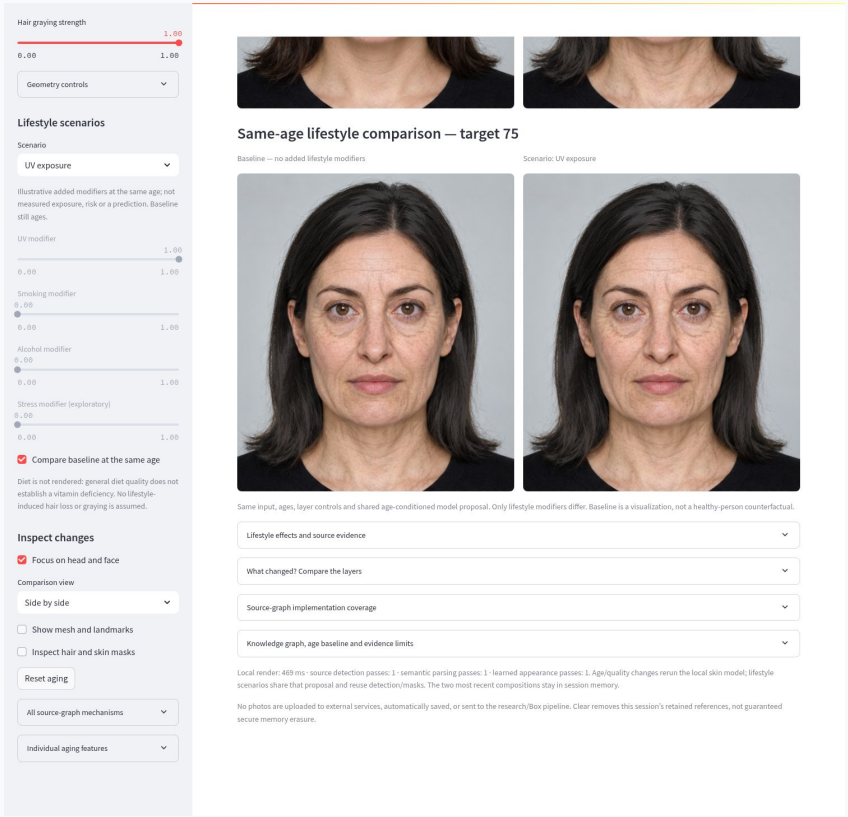
Heavy unhealthy



Light unhealthy

CURRENT AVATAR





SOFTWARE BREAKDOWN

FACE SHAPE

Mesh-based facial warping

Uses facial landmarks and geometric transformations to simulate age-related changes in facial structure, including sagging and volume loss.

SKIN & HAIR

AI-assisted skin & hair aging

Applies a local age-conditioned neural model for wrinkles and skin texture, combined with facial segmentation for hair graying.

LIFESTYLE

Knowledge graph-driven rendering

Maps aging mechanisms and lifestyle factors to controllable facial features, enabling localized adjustments and scenario comparisons.

HOW AVATAR WORKS

From selfie to reflection in five steps

1

Input

Take a selfie and enter age, demographics, and lifestyle

2

Generate

Identity-preserving AI creates an aged portrait based on biomedical data

3

Explain

See which features changed and the biology behind them

4

Compare

Change a habit and view an alternative future

5

Reflect

Connect today's choices with long-term health

ETHICS AND APPROVAL

IRB submitted

October 13, 2026

What is an IRB?

An **I**nstitutional **R**eview **B**oard is required to review research involving human subjects to ensure participants are treated safely, fairly, and with respect.

OVERVIEW

Design	Minimal-risk pilot study
Participants	Up to 100 UIUC students, 18+
Session	One in-person session, about 35 minutes
Training	All team members CITI-certified
Privacy	Participants assigned random ID with all data stored in a restricted UIUC Box folder

WHAT WE'RE MEASURING

Three questions for the pilot

Realism

How authentic does the aged avatar feel?

Emotion

How do participants feel when they see their older selves?

Usability

What should change in the design for the next version?

FUTURE PLANS

Looking Ahead

Pilot Study	Run a pilot with up to 100 UIUC students and gather feedback
Transparency	Explain each aging feature: what changed, what habit caused it, and why
Improvements	Improve image with study feedback, new research, and better AI models
Share the Results	• Undergraduate Research Symposium (April 2027) • National Conference on Undergraduate Research (April 2027) • International Journal of Undergraduate Research and Creative Activities

Future Plans

Share the Results

Undergraduate Research Symposium

WHEN

April 28–29, 2027

WHERE

Illini Union, Urbana, IL

NEXT STEP

Prepare a 200-word abstract and apply to present.

NCUR

National Conference on undergraduate research

WHEN

April 12–14, 2027

WHERE

Grand Rapids, MI

NEXT STEP

Submit the AVATAR abstract by December 4th, 2026.

IJURCA

International Journal of Undergraduate Research and Creative Activities

WHEN

Spring 2027

NEXT STEP

Analyze the pilot data and draft the manuscript.

SIGNIFICANCE

AVATAR makes the link between today's habits and the future visible, realistic, and understandable.



Inclusiveness

Represents aging across genders, ethnicities, and skin tones



Transparent

Explains which features changed and why



Holistic

Shows how genetics, environment, and habits combine



Empathy-Building

Helps destigmatize aging and connect with older adults

THANK YOU

Questions?

FULL REFERENCE LIST

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